School of Computer Science and Engineering

Faculty of Engineering

The University of New South Wales

Efficient Grasp using Superquadrics for Human Service Robots

by

Jayden Kanbour

Thesis submitted as a requirement for the degree of

Bachelor of Engineering in Computer Engineering

Submitted: November 2025

Supervisor: Claude Sammut

Student ID: z5316799

Abstract

To serve their purpose, human service robots need a grasp solution that enables them to reliably and quickly perceive and interact with objects in their environment, while adhering to the constraints of their hardware and software. In recent years, deep learning-based grasping solutions have taken the spotlight for their broad applicability and high accuracy. However, to achieve this, they required high computational costs, long training times, and a reliance on large annotated datasets. In contrast, geometric approaches such as superquadrics offer lightweight representations but struggle with partial views, occlusion, and complex object geometries. In light of recent research, this thesis presents a three-stage pipeline that integrates base component identification, superquadric fitting, and geometric grasp selection to generate efficient and reliable 6D grasp poses from a single segmented depth image. The system does not require prior object models, training data, or multi-view reconstruction, and is designed for seamless integration into existing robotic perception pipelines. Evaluation demonstrates that the proposed method achieves significantly higher computational efficiency than state-of-the-art learning-based approaches, with a runtime of 0.018 seconds and the lowest memory usage among all tested systems. Accuracy tests using the Toyota HSRb show competitive grasp success rates in isolated and cluttered scenes, though performance decreases under heavy occlusion due to single-view limitations. Overall, the results highlight the potential of adaptive superquadric-based modelling as a fast and resource-efficient alternative to deep learning for robotic grasping, particularly for real-time operation on hardware-constrained service robots.

Acknowledgements

I would like to acknowledge and show my appreciation to all those who make this possible. Firstly, I would like to thank God for giving me Grandparents who sacrificed everything they knew to come to a country that would provide their children and grandchildren with opportunities unimaginable to them. I am grateful for my parents, who continue to sacrifice all that they have to enable me and my siblings to pursue any endeavour that I put our minds to. Both of you are my inspiration and best friends, and I could not be more thankful for both of you. I am also grateful to my friends and family for providing me with patience, love, and enduring support. This great achievement reflects the generosity and love you have provided me. I would also like to thank my partner, Jamelle. Your unwavering support and faith in me have allowed me to pursue ambitions that I never thought I was capable of. Lastly, I would like to thank Claude Sammut and the rest of the 2025 rUNSWift team for your guidance through this entire process. It would not be possible without you all.

Contents

List of Figures

Chapter 1

Introduction

Robot vision for object manipulation remains a longstanding challenge, particularly in unstructured and unfamiliar environments. The ability of human service robots, such as the Toyota HSR, to perceive objects in its surroundings directly impacts how safely and effectively it can interact with them. Recent advances in machine learning have significantly contributed to solving this problem, most notably through end-to-end solutions. However, these solutions require substantial data ingestion and computational resources to run effectively. Furthermore, while many methods can perform in an unknown environment with new objects, they still rely heavily on extensive training to achieve efficient performance.

While recent advancements in vision pipelines have primarily focused on deep learning and data-driven approaches, compact geometric methods like superquadrics, have long been of interest due to their low computational cost, fast processing, and robustness. However, existing implementations of superquadrics often struggle when the 3D data is incomplete, such as in the presence of noise and heavy occlusion, and frequently fail when only one perspective of the object is available. Additionally, although they perform well when modelling simple shapes such as cuboids, cylinders, and spheres, they struggle to represent complex or concave objects accurately.

This thesis hypothesises that a pipeline stemming from superquadric and geometric-based grasping can perform accurate grasping of unknown objects more efficiently than state-of-the-art machine learning, while addressing the current limitations of existing superquadric-based systems. To investigate this hypothesis, the proposed method introduces a three-stage grasp estimation pipeline that operates solely on a segmented depth image and requires no prior object models or training data. The system first extracts a clean point cloud of the target object and employs a RANSAC algorithm to break it down into simple base components. Each component is then fitted with a superquadric representation, providing stable shape, curvature, and size information. Finally, the system evaluates these representations to identify collision-free, force-closure grasp points and generates the corresponding 6D grasp poses. This lightweight architecture is designed for seamless integration with existing vision modules, minimal computational overhead, and reliable performance on previously unseen objects.

The remainder of this paper is structured as follows. Chapter 2 presents a summary of the background literature relevant to this thesis. Chapter 3 describes the proposed method in detail, outlining the requirements and processes of the three-stage pipeline. Section 4 evaluates the system's efficiency and accuracy, comparing it with baseline and state-of-the-art approaches. Finally, Section 5 discusses the results, identifies limitations, considers practical applications, and highlights avenues for future work.

Chapter 2

Background

2.1 Object Component Breakdown

Breaking down complex objects into their basic geometric components has become an essential strategy in classic robot manipulation. It assists robots in generating possible grasps for complex or large objects by deriving necessary information from the geometric properties of simpler shapes that comprise them, such as cubes, cylinders, and spheres.

Significant research demonstrates the benefits of extracting the object's basic geometric components for grasping. Early work on RANSAC-based approaches by Krainin et al. (2011)[11], Maitin Shepard et al. (2010)[12], and Richtsfeld et al. (2012)[16] shows that fitting planes, cylinders, and spheres directly from depth data can support reliable grasp selection. However, these approaches often rely on costly search algorithms and perform inadequately when objects are obstructed or in poor lighting.

More recently, studies using superquadrics have explored a range of techniques for achieving higher accuracy. Goldfeder (2010)[7] demonstrates the use of decomposition trees, in which each branch divides the target object into two parts until each component can be easily represented as a superquadric. Alternatively, Xun Tu et al. (2025)[19]

proposed that a marching primitives approach can decompose complete 3D models by segmenting the mesh into multiple regions and framing the decomposition as a nonlinear least-squares optimisation task. Despite this, these approaches heavily relied on existing watertight meshes of the target object, limiting their practical use in unknown settings.

The current paper addresses the noted limitations by employing a lightweight, adaptive method based on the classic RANSAC approaches, further adapted for superquadric modelling to enable real-time calculation from partial RGB-D images. The system further builds on classic approaches by introducing greater tolerance for the depth and estimated surface normals of each point, thereby better recognising curved surfaces. Further enabling efficient and reliable detection of basic components while maintaining the flexibility needed for unseen objects.

2.2 Superquadrics

Superquadrics are a family of parametric shapes that provide a compact, flexible representation of a wide range of 3D geometries. Introduced into the field of computer vision by Alan H Barr (1981)[2], they encompass forms such as superellipsoids, supertoroids, and superhyperboloids. In model fitting, a superquadric is adjusted to a 3D point cloud by estimating shape and pose parameters that minimise the distance between the surface and observed data points. The most common representation of a superquadric is defined by the inside–outside function[1]:

$$F(x,y,z) = \left(\left| \frac{x}{\alpha_1} \right|^{\frac{2}{\varepsilon_2}} + \left| \frac{y}{\alpha_2} \right|^{\frac{2}{\varepsilon_2}} \right)^{\frac{\varepsilon_2}{\varepsilon_1}} + \left| \frac{z}{\alpha_3} \right|^{\frac{2}{\varepsilon_1}} \tag{2.1}$$

Where the variables α_1, α_2, α_3 control the scale along the x, y, and z axes, and ϵ_1, ϵ_2 control the shape's roundness. While superquadrics offer a compact and efficient way to model objects, they are sensitive to initial parameter estimates and may struggle with non-convex or highly irregular shapes. The accuracy is further reduced in the presence of noise or partial data, making robust initialisation and outlier removal necessary for real-world applications [3].

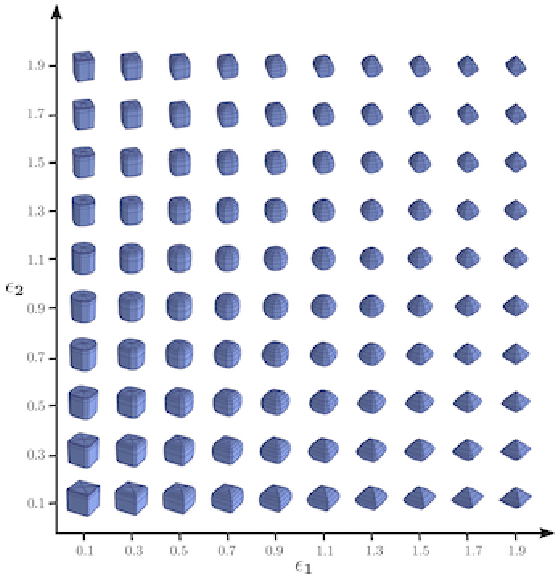

Figure 2.1: Superquadric Shape Space: The figure visualises superquadrics with different variables [1]

In robotic grasping, superquadrics facilitate object handling due to their simple, symmetrical shapes, which can be described by a few parameters. Wu et al. (2023)[19] proposed a method that does not require training and fits several hidden superquadrics to point cloud data by separating unusual points. The advantage of their method allows for reliable 6D grasping without training, but requires cameras to remain fixed and struggles with thin or messy groups of objects. Xun Tu et al. (2025)[19] proposed SuperQ-GRASP, which decomposes complete 3D models into a mosaic of superquadrics to find feasible grasps for large objects. This, however, is dependent on an existing complete reference model and cannot handle very fine features. Notably, Dunes et al.(2008)[5] and Marton et al. (2010)[13] took advantage of the symmetrical nature of superquadrics to estimate fesabel grasps from single images, but require the objects to be symmetrical across at least one axis. Whereas, Vezzani et al. (2018)[21] described both the object and the robotic gripper using superquadrics to speed up grasping; however, their method requires solid, unchanging shapes.

As is evident, superquadrics present a lightweight, model-free alternative to deep learning approaches, aligning well with the need for fast, robust perception pipelines in robotics. While they provide a compact and efficient representation for many primitive objects, they struggle with incomplete models and often fail to represent concave or irregular shapes. To address these limitations, the thesis proposes fitting superquadrics not to the entire object, but to individual graspable surfaces. By decomposing the object into planar or near-planar regions before fitting, the method reduces the model complexity each superquadric must explain, further supporting low cost and real-time processing.

2.3 Grasp Pose Estimation

Grasp estimation predicts where and how a robot should position its gripper to stably and efficiently grasp an object. Efficient and accurate grasping remains a central challenge in robotic manipulation, particularly in unknown environments where robots must operate with partial or noisy data. Early analytical methods are grounded in the concepts of form closure and force closure[4], which define the conditions for mechanical stability. Form closure ensures that an object is fully constrained through contact geometry alone, while force closure considers the balance of external forces and torques applied to the object. These formulations provide a rigorous theoretical foundation for grasp stability.

Classic geometric approaches use an object's geometry and appearance to generate large sets of grasp hypotheses from 3D data and to evaluate possible grasps using selected criteria stemming from the principles of force and form closure. A clear example conducted by Pas and Platt (2015)[15] proposes a sampling-based method that uses surface normals and collision constraints to identify antipodal grasps, demonstrating strong performance in cluttered scenes. Similarly, Zapata Impata et al. (2019)[22] developed a fast, model-free approach that derives grasp candidates from single viewpoint clouds by analysing local curvature and centroid distance. While efficient, these methods remain sensitive to occlusion, point sparsity, and irregular object geometry, limiting their application to deformable or highly complex shapes.

In recent years, there has been a transition toward learning-based grasping, enabling perception systems to infer grasp candidates for both known and unknown objects. These methods have achieved strong results in previously challenging scenarios, including cluttered environments, partially obstructed objects, and objects in unpredictable or unconventional poses. State-of-the-art solutions such as GraspNet-1billion[6] and NVIDIA's DOPE[18] achieve real-time grasping, demonstrating the impressive performance achievable with data-driven approaches. Furthermore, these deep learning approaches are supported by research into large-scale datasets such as LINEMOD[9], which focuses on occluded and irregular shapes, and HOPE[20], which captures household and office objects under varied lighting. Despite these results, significant barriers remain for deployment in human service robots, as learning-based pipelines often require substantial computational resources, long training times, and large annotated datasets. Additionally, the time and data requirements for retraining or fine-tuning models are not feasible in rapidly changing or unstructured environments.

The paper proposes an alternative solution that addresses several of the noted limitations by reducing the grasp estimation process to a purely geometric one. By leveraging superquadric representation rather than heavy learning-based models, the aim is to enable grasp estimation that generalises across object types, environments, and levels of observability without requiring extensive training data. In doing so, this approach seeks to support lightweight, real-time, and resource-friendly grasp pipelines that are suitable for real-world assistive and service robotics.

Chapter 3

Solution

The proposed solution is a three-stage pipeline that generates feasible grasp poses for unknown objects using only a segmented depth image. It is designed for seamless integration with the robot's existing vision and manipulation modules and requires only a depth camera, a robot arm capable of executing a force closure grasp, and either a segment mask or a bounding box produced by any onboard vision model. The system performs all processing internally and makes no assumptions about object geometry or prior training, prioritising minimal compute in terms of memory, time, and processing cost. An overview of the pipeline and its data flow is shown in Figure 3.1, and the complete source code and demonstration videos are available in the GitHub repository, see Appendix A.1.

The pipeline operates in three stages.

1. Object point cloud extraction and component identification converts the segmented depth data into a clean point cloud and isolates the basic geometric components suitable for grasp reasoning.

2. Superquadric fitting represents each region with a compact analytic shape to provide stable surface normals, curvature, and size information, to minimise the search required to identify a suitable grasp.

3. Grasp point filtering and pose generation evaluates each fitted superquadric to identify collision-free, force-closure grasp points and computes a corresponding 6D grasp pose for the robot gripper as well as an optional 2D pose for the robot's base to best position itself.

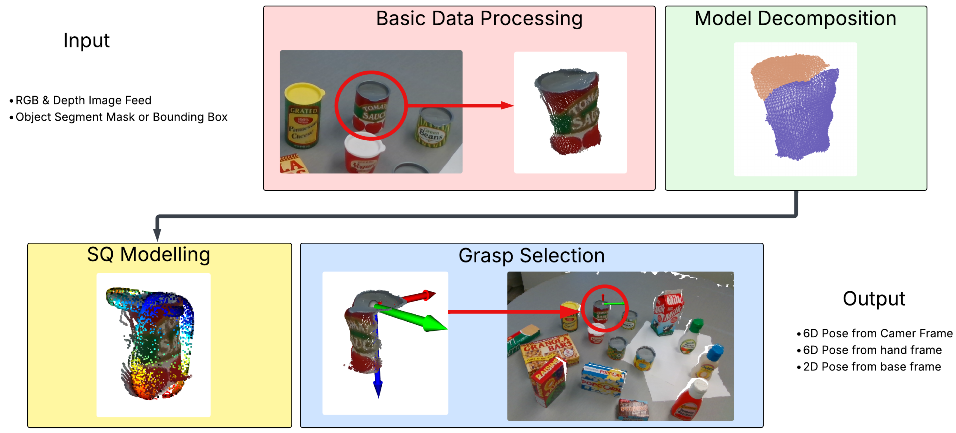

Figure 3.1: Pipeline of proposed solution. Objects from HOPE dataset [20]

3.1 Object Point Cloud Extractions and Component Identification

At the initial stage of the pipeline, the primary objective is to isolate the target object from its environment and subsequently decompose it into its smaller base components. The required input includes the RGB and depth images, as well as a bounding box or segment mask to identify the target object. The solution requires the robot to be running a vision model capable of object detection. This initial stage is significant, as it reduces the search space and the computational cost required to detect feasible grasps in later stages. Unlike many existing approaches, the proposed method does not attempt to estimate or reconstruct the complete 3D model of the object using references, prior knowledge, or multiple viewpoints. Instead, it relies solely on information from a single frame of reference, making it suitable for applications with limited data or sensing constraints.

To achieve base component decomposition from a single view, a RANSAC-based plane detection algorithm was implemented. Built on top of classic RANSAC approaches, this algorithm is augmented with a high tolerance for depth and normal variation and a low tolerance for point and point cluster distance, enabling the recognition of smooth or gently curved surfaces that may otherwise be overlooked by standard segmentation routines. The algorithm found in appendix A.3 samples small triplets of points that satisfy geometric consistency conditions, fits a provisional plane through these seeds, and then classifies inliers using combined distance, normal, and depth constraints. The best supported plane is extracted and removed from the cloud, and the process repeats until no further valid surfaces remain. Thus, providing a stable, noise-resistant method for isolating base-object components before superquadric fitting.

Figure 3.2 shows how this method performs for objects of varying complexity. As observed, it works well for objects with clearly defined segments and planes, such as cans and boxes, as well as complex objects with several non-primitive components. However, because the algorithm relies solely on geometric information, it cannot distinguish components with non-geometric divisions, such as the smooth transitions shown between the dragon's body and wings in figure 3.2.

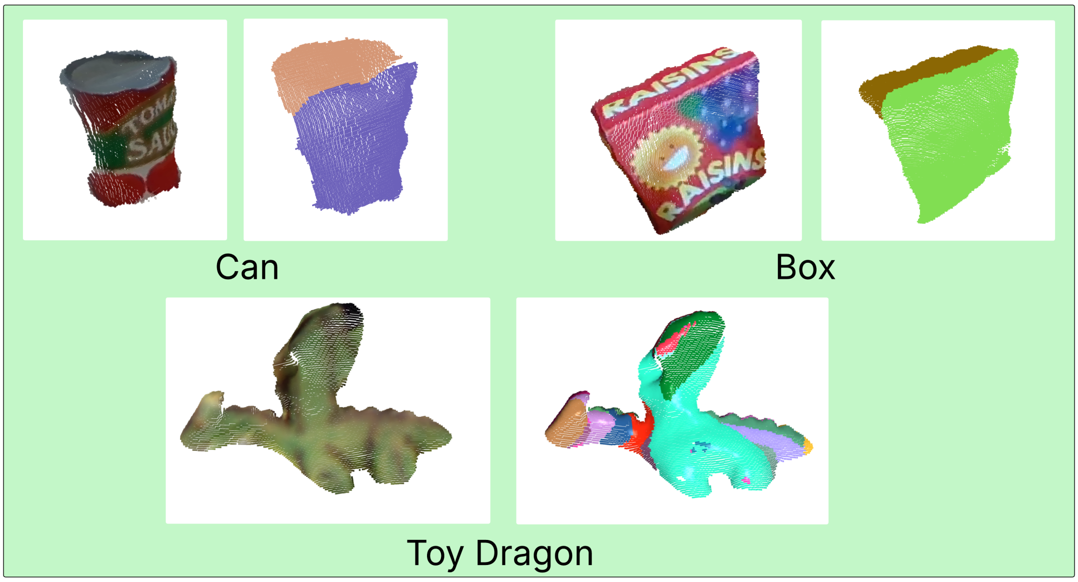

Figure 3.2: Sample of base component decomposition. Objects from TUD-L[10] and HOPE[20] datasets

While this method is relatively simple and computationally efficient, it is also less precise than more advanced techniques such as Marching Primitives or Expectation-Maximisation. Unlike those model-based or probabilistic methods, the RANSAC approach does not depend on a complete object model or require complex statistical inference, allowing it to operate effectively with minimal data and processing requirements. This trade-off between accuracy and efficiency makes it a practical first step toward reliable object segmentation and analysis in single-view depth data.

3.2 Superquadric Fitting

The goal of the second stage is to fit an appropriate superquadratic to each previously identified base component, such that the system can represent each component as a compact mathematical model. By doing so, the system can infer possible grasps from the inherent symmetry and antipodal properties of the superquadrics, thus minimising the computational effort required to identify feasible grasps in the final stage. The approach is inspired by the work of Xun Tu et al. (2025)[19] on Super-Q Grasp, where they represented large objects as chunks of graspable superquadrics. The present paper extends their research by applying it to small tabletop objects and by adapting it to work with partial data rather than a complete model.

To reiterate, a superquadric can be formulated using six parameters. Three scalar parameters α_1, α_2, α_3 govern the shape's span in each coordinate direction (x, y, z). Two shape parameters ϵ_1 and ϵ_2 dictate the model's shape and roundness along the vertical and horizontal planes. In this work, the scalar parameters α_1, α_2, α_3 are derived directly from the geometric span of each base component along each corresponding plane, after outlier removal to prevent overestimation.

Interestingly, the shape parameters ϵ_1 and ϵ_2 are estimated statistically from the point distribution within each base component using a kurtosis-based approach as demonstrated in algorithm 1. The point cloud is first centred to remove spatial bias, and kurtosis is computed along the x, y and z axes to understand how tightly points cluster around the mean. Higher kurtosis values indicate longer or narrower surfaces, whereas lower values represent flatter, more uniform surfaces. These measurements are linearly mapped into a bounded range to maintain stability.

Algorithm 1 Estimate Superquadric Shape Parameters e_1 and e_2

function ESTIMATEE(*pcd*)
  points ← point coordinates from *pcd*
  center ← geometric center of *pcd*
  centeredPoints ← points minus center
  Compute Fisher kurtosis vector *krt* of centeredPoints along x, y, z axes
  $k_z \leftarrow krt[2]$ ▷ kurtosis along z axis
  $k_{xy} \leftarrow (krt[0] + krt[1])/2$ ▷ average kurtosis in xy plane
  $e_1 \leftarrow \text{clip}\,(1 + (k_z - 3) \cdot 0.1,\ 0.3,\ 2.0)$
  $e_2 \leftarrow \text{clip}\,(1 + (k_{xy} - 3) \cdot 0.1,\ 0.3,\ 2.0)$
  return (e_1, e_2)
end function

By fitting superquadrics to base components as opposed to the complete model, the system addresses the challenges of modelling non-convex and irregular objects. As illustrated in figure 3.3, the mug is decomposed into two superquadric components rather than being approximated as a single cylinder. A generic superquadric fitting pipeline would typically model the entire cup as one cylindrical shape, which would make it difficult to identify feasible grasps from the rim without additional processing, such as model subtraction or component recognition. By isolating only the graspable surfaces, this method generates two clear grasping opportunities along the rim and several additional options along the front-facing surface of the mug.

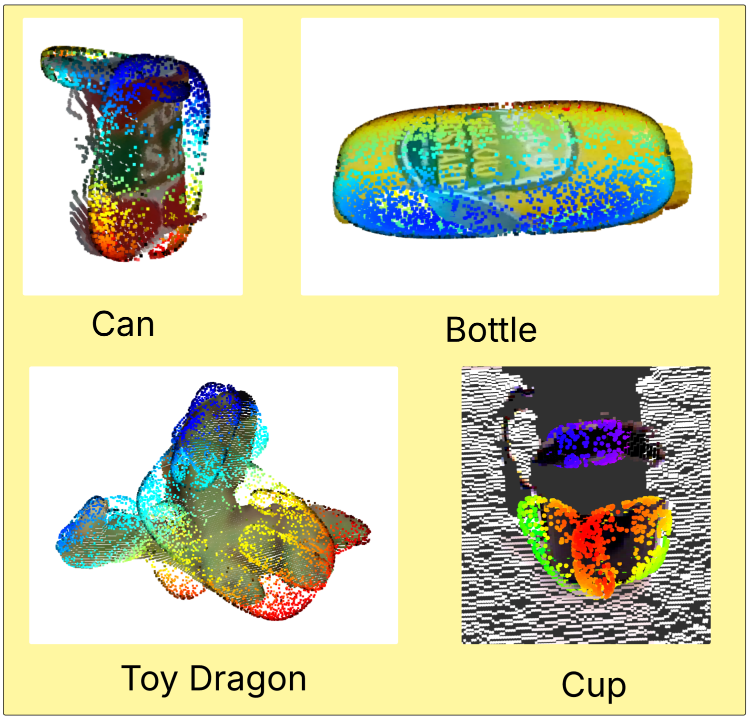

Figure 3.3: Sample of Superquadric fitted to various objects

Therefore, the proposed approach to superquadric fitting demonstrates its ability to handle incomplete and partial data, as well as to capture the features of non-convex and irregular shapes necessary for grasping. Furthermore, the use of readily available information, including the base components centre, point span and orientation provides a lightweight, data-driven mechanism for rapid and robust estimation of graspable surfaces as superquadrics. However, as superquadrics can not precisely replicate the object's shape, they increase the risk of inaccurate grasp opportunities in the final stage.

3.3 Grasp Pose Selection

The objective of the grasp pose selection stage within the grasp planning system is to determine the optimal 6D Pose (point and orientation) for the robot to grasp the object securely. At this stage, the system also considers a desired grasp direction supplied by the calling process, which can be specified in general terms such as top, front, left, or right. Poses that align with this desired direction are prioritised during the search process. The selection process comprises two steps:

1. Step one, the system identifies the most suitable superquadric (result of the first two stages) that aligns with the desired direction.
2. Step two, the system determines the precise point and orientation on the selected superquadric's surface that provides a stable and feasible grasp.

3.3.1 Base Component Selection

The purpose of the base component selection step is to choose which superquadric segment of the overall object offers the most opportunity for a suitable grasp, given the robot's mechanical constraints and the requirements of the overall task. To achieve this, the system evaluates each superquadric with close attention to its centre, orientation, axis lengths and surface normals. Thus, the system can quickly infer whether the robot's gripper can physically reach and enclose that component of the object. Two considerations to complete this step include:

- Superquadric Position: The relative positions of the centre and surface normals for each superquadric are examined to determine whether it exposes a graspable surface from the requested direction. For instance, if a top grasp is asked for, the candidate superquadrics must present an upper-facing surface with sufficient clearance. If none are present, then the algorithm may need to select an alternative grasp direction.
- Superquadric Size: The size of each superquadric is compared to the robot's maximum and minimum gripper opening widths. It ensures that the component is neither too thick to grasp nor too thin to provide stable grasping.

By combining these checks, the system guarantees that the selected component not only aligns with the intended grasp direction but also represents the most promising region of the object for generating an antipodal, force-closure grasp. This filtering stage significantly reduces the computational load on the subsequent step by eliminating base components that cannot support a stable grasp.

3.3.2 Surface Point and Orientation Selection

Once the appropriate superquadric has been selected, the next stage identifies a precise 6D pose on its surface that satisfies a set of grasping criteria. These criteria include:

- Ensuring that a grasp at that point allows for a force closure
- Verifying that the local width of the object at that point lies within the mechanical limits of the robot's gripper
- Confirming that the grasp orientation at the chosen point aligns with the desired direction.

To achieve this, the system draws inspiration from previous work by [21], who limits candidate points to those lying along the superquadric's primary axes as observed in 3.4. Due to the inherent symmetry of superquadrics, any point sampled along a primary axis naturally corresponds to antipodal surface points on the adjacent side of the object. This guarantees the first requirement for stable grasping while reducing the number of points that must be evaluated.

Furthermore, sampling along these axes provides an efficient means of determining the local width and thickness of the object at each candidate point. By comparing this to the robot's maximum and minimum gripper size, the system can immediately eliminate infeasible grasp points. Determining the grasp orientation at the chosen point then becomes straightforward. As illustrated in figure 3.5, the reference pose for each gripper aligns the x, y, and z axes with the red, green, and blue directions, respectively. In setting the orientation for the grasp pose, the z axis is always set first and is directed towards the centre of the superquadric, ensuring that the approach vector points into the object. The x-axis is then selected to span a region of the surface where the local

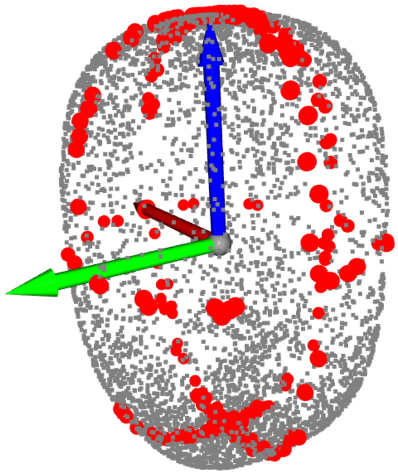

Figure 3.4: Basic superquadric with primary points highlighted

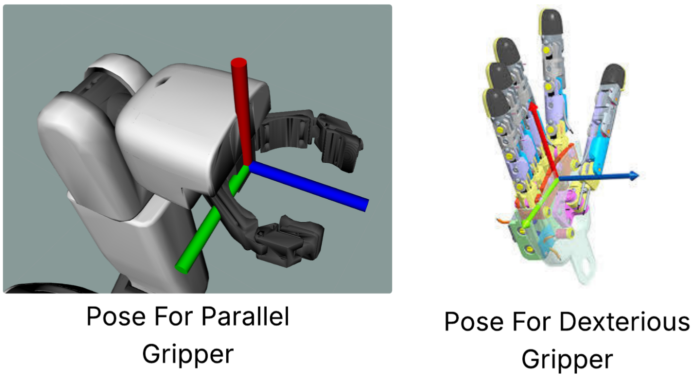

Figure 3.5: Poses For Different Robot Grippers. Dexterous gripper from [21]

thickness is suitable for a secure grasp, enabling either force or form closure. The y-axis is defined automatically. As shown in figure 3.6, this method performs robustly across diverse objects, including both tabletop items and irregularly shaped objects. Its success stems from the preceding pipeline stages, which decompose complex objects into graspable superquadric segments and use the inherent symmetry and geometric structure of superquadrics to simplify grasp evaluation. Together, these components enable efficient, real-time grasp estimation with substantially reduced computational demands.

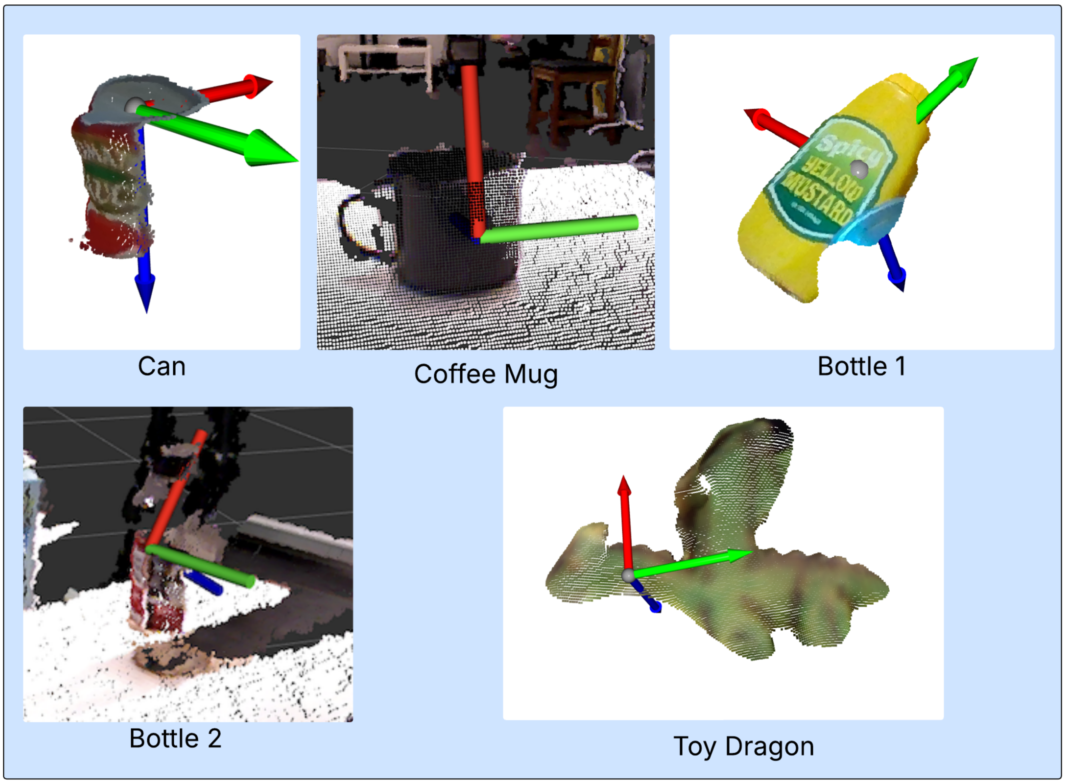

Figure 3.6: Estimated 6D Poses for a range of objects

Chapter 4

Evaluation

When evaluating the system, both efficiency and accuracy are assessed through a range of quantitative measures. Efficiency is evaluated in terms of processing time, computational load, and memory usage required to calculate a successful grasp. Accuracy, on the other hand, is measured by the ratio of successful grasps achieved across various test scenarios and object types. To ensure the objectives of this thesis are met, the proposed solution is compared against a selection of existing benchmarks and state-of-the-art approaches, including GraspNet-1billion[6], DOPE by NVIDIA[18], and GPD[17]. Furthermore, to assess the proposed solutions' improvements over the limitations of existing superquadric approaches, isolated testing was conducted on a range of object types.

4.1 Efficiency tests

The efficiency tests were conducted entirely in simulation to ensure controlled and repeatable conditions. Each system under evaluation was executed with only the bare minimum dependencies and processes required for operation, thereby isolating the pose estimation module's performance from external computational influences. The standardised HOPE dataset[20] was used for all training and testing, as it comprises a set of tabletop objects commonly found in a home environment. For each target object, the system generated a corresponding grasp pose prediction, allowing for precise measurement of computational performance. Throughout the process, three key performance indicators were monitored: processing time, CPU utilisation, and memory consumption. Processing time was recorded as the total duration required for the system to generate a complete pose estimation from the moment of input reception. CPU utilisation, tracked across multiple cores, was used to assess the computational load imposed on available processing resources, providing insight into scalability and runtime efficiency. Memory consumption was measured to evaluate the system's resource usage and its potential suitability for deployment on platforms with limited hardware capabilities. To enhance the reliability of results, each test was repeated multiple times under identical simulation conditions, and the collected metrics were averaged to minimise the influence of outliers. These efficiency measurements, together, form a comprehensive profile of the system's computational performance, serving as a foundation for comparison with existing benchmarks and alternative implementations. See Appendix A.2 for a full list of the computer setup used for testing.

Table 4.1: Performance comparison of different grasp estimation systems

System	**Time(s)**	**Memory(Mb)**	**CPU(%)**
This paper	**0.018s**	**78Mb**	83%
GPD	0.39s	347Mb	**55%**
DOPE	0.053	1213Mb	122%
GraspNet	0.08s	1804Mb	100%

4.2 Accuracy tests

Accuracy testing was performed as two main components. The first measured grasp performance across different placement conditions, including isolated, cluttered, and obstructed scenes, and was compared to state-of-the-art solutions. The second assessed grasp success for different object categories, namely primitive, concave, and complex shapes, and was used to assess this solution's improvements on the limitations noted in other superquadric-based manipulation pipelines. Each experiment was repeated 5 times, providing a reliable indication of how the proposed method performs across varied manipulation contexts and its potential applicability in unstructured environments.

4.2.1 Accuracy Test 1: Testing Grasping in Different Scenarios

The first set of accuracy tests explored grasp success across three distinct scenarios, each designed to test different situations a human service robot might encounter. In the standalone object scenario test, individual objects were placed in isolation to assess the system's baseline performance under ideal conditions. The cluttered environment scenario introduced multiple non-obstructive objects, testing the system's ability to identify and isolate target items within visually complex scenes. Finally, the obstructed-grasp scenario introduced partial visual and physical obstructions caused by other objects. Model training and testing were again conducted using the HOPE dataset[20], and results were compared with noted baseline and state-of-the-art solutions.

Table 4.2: Qualitative grasp success rates across different systems

System	**Isolated**	**Cluttered**	**Occluded**
This paper	4/5	4/5	2/5
GDP	4/5	3/5	3/5
DOPE	**5/5**	**5/5**	3/5
GraspNet[1]	**5/5**	**5/5**	**4/5**

4.2.2 Accuracy Test 2: Testing Grasping for Different Object Types

The second set of accuracy tests was used to assess whether this solution has improved upon the limitations addressed by similar superquadric-based solutions. Notably, the inability to estimate grasps for partial models, concave objects, and complex objects. To achieve this, the solution was tasked with successfully grasping three distinct objects

[1]Due to hardware and software incompatibility, the GraspNet-1billion system could not be evaluated on the Toyota HSRb. Reported results are taken from the original paper [6] and survey [14].

from a single perspective: a can as a baseline test, a coffee mug from the rim to represent concave objects, and a chair to represent complex objects.

Table 4.3: Qualitative grasp success for a range of object types

Primitive (can)	**Concave (Mug)**	**Complex (Chair)**
5/5	4/5	4/5

Chapter 5

Discussion

5.1 Analysis

The evaluation conducted in chapter 4 concludes three things:

- The proposed system is capable of generating highly efficient grasps, especially compared to deep learning approaches such as GPD, DOPE, and GraspNet. Table 4.1
- The system's grasps are passable but cannot match the performance of deep learning methods. Table 4.2
- The system can address the limitations set by other superquadric-based grasp pipelines. Table 4.3

The efficiency test results highlight a clear advantage of using a superquadric-based approach over a deep-learning method. It produces significantly faster processing and reduced memory requirements. With an average runtime of only 0.018 seconds per prediction, the system is significantly faster than all competitors, including GPD, which takes more than 20 times as long. The lowest recorded memory consumption of all evaluated systems makes the method particularly well-suited to platforms with constrained hardware, such as the Toyota HSR. Although CPU utilisation is moderately high, it remains well within acceptable bounds and significantly lower than DOPE and GraspNet, whose processing demands exceed the limits of many smaller robots. Overall, these results indicate that the superquadric-based pipeline is both lightweight and responsive, enabling near real-time deployment without specialised hardware acceleration.

The first set of accuracy tests reveals a more confronting comparison. In isolated and cluttered conditions, the proposed solution achieves a strong success rate of 4 out of 5 grasps, almost matching the performance of DOPE and GraspNet. The most challenging scenario, occluded grasps, shows a further decrease to 2 out of 5 successes. The decline in performance under visual obstruction is consistent with the limitations of single-view perception, in which partial point cloud information and a lack of context about the object and its environment affect the precision of fitted superquadrics. The failures occurred during tests in which the target objects were heavily obscured from multiple directions. While this remains respectable given the limited view of the object, it falls short of benchmark methods that rely on more data-hungry or fully supervised deep learning strategies. Despite these differences, the results highlight a clear trade-off between efficiency and accuracy. Methods such as DOPE and GraspNet achieve higher grasp success rates, particularly under obstructions, but require substantially greater computational resources, longer prediction times, and, in some cases, significant GPU support. In contrast, the proposed approach achieves competitive accuracy while operating at a fraction of the computational cost. This makes it highly practical for real-world applications in human service robotics, where reliability must be balanced with strict hardware and software limitations and the need for responsive behaviour.

Further accuracy testing was conducted to evaluate how well the proposed system was able to overcome the limitations raised by previous superquadric-based grasping methods. A key contribution to achieving the result is the pipeline design choice to model superquadrics and generate grasp estimates not over the entire object, but over its individual graspable components. As a result of this component-based modelling strategy, the system demonstrated the ability to grasp objects from partial views, concave objects, and irregular or complex objects. These improvements highlight the advantage of focusing on local graspable surfaces rather than approximating the entire object with a single superquadric. Consequently, the method extends the application of superquadric-based grasping to a broader class of objects and enhances its suitability for real-world manipulation tasks with limited visibility and unpredictable object geometry.

Overall, the evaluation indicates that the system fulfils its intended design goals: it provides fast, resource-efficient grasp estimation with accuracy suitable for everyday manipulation tasks in household or care environments. It offers a promising alternative to heavier learning based architectures, particularly in scenarios where speed, interpretability, and deployability on modest hardware are prioritised.

5.2 Limitations

Despite the system's high efficiency results and ability to work with a range of unknown objects, several factors limit its reliability and accuracy in complex or dynamic environments.

Firstly, the system's lack of contextual understanding of the target object, the environment, and the task being executed makes it difficult to determine whether a grasp will be successful, safe, and efficient. To partially address this, the system accepts additional metadata, including the object's tracking ID and name, to ensure the correct object is targeted. Similarly, supplying a desired grasp direction, such as top, front, or side, allows the user or a higher-level planner to impose safety or task-driven constraints on the grasp approach. However, these additions do not fully compensate for the lack of higher-level task awareness provided by deep learning approaches.

Secondly, the grasp planning system is dependent on an external vision module for object detection and segmentation. As a result, the overall performance and reliability of the grasp are tightly coupled to the quality of the running vision model. Factors such as lighting variations and sensor resolution can impact segmentation accuracy and directly affect the quality of the resulting grasp estimation.

Finally, the model decomposition stage relies on variations in surface depth and curvature to segment the object into meaningful base components. Although this method performs well for most objects, it struggles when the object's components have smooth or gradual transitions. In such cases, the system may incorrectly merge separate components or fail to identify critical visual distinctions, such as changes in colour. The limitation is therefore closely linked to the system's lack of contextual knowledge, as it cannot infer component boundaries based on object semantics or expected functional structure.

5.3 Practical Application

As discussed in the previous section, the developed system demonstrates strong performance in terms of computational efficiency, processing time, and resource usage. Such characteristics make it well-suited for integration into robotic systems that are already operating under heavy computational loads. However, despite these strengths, the system has several limitations that restrict its reliability for fully autonomous or high-stakes object manipulation tasks.

Accordingly, it is recommended that the system be employed as a backup grasping module within service robots. While deep learning based approaches typically offer superior accuracy, contextual understanding, and adaptability in complex real-world environments, they are often computationally expensive and require lengthy introductions to new objects. In scenarios where deep learning systems are unable to perform in cases where the robot's computational resources are already heavily utilised, or when the primary perception pipeline fails, the proposed system can act as a lightweight fallback. Its low latency and minimal processing requirements allow the robot to continue operations while generating a feasible grasp pose.

Beyond its application in human surface robots, the proposed method can also serve as a supporting component for deep learning based frameworks. In this role, it can provide accurate initial pose estimates that reduce the search space for learning-based algorithms, thereby accelerating convergence and improving inference efficiency. Additionally, it may be used as a low-cost data generation during training, providing reliable grasp candidates that help refine model performance across diverse scenarios.

Overall, while the system may not yet achieve the adaptability of data-driven approaches, its speed, simplicity, and robustness make it a valuable module for enhancing the resilience and computational efficiency of robotic manipulation pipelines.

5.4 Future Work

Given the performance of the proposed system and the limitations identified throughout the thesis, several avenues for future work are proposed that could either extend the current proposal or establish new research directions.

The first and most immediate extension should address the system's limited contextual awareness of objects and their surrounding environment, identified as the first limitation. Enhancing this aspect would require incorporating additional information from the existing vision model or from the world model. Inputs such as object class, estimated dimensions, precise orientation, and the presence of nearby obstacles would improve the system's ability to compute accurate grasps in cluttered and occluded scenes. Such contextual data would reduce the likelihood of unreliable or unsafe grasps. However, integrating this additional information introduces new challenges, particularly the need to preserve lightweight computation and real-time performance while handling more complex reasoning.

A second direction involves expanding the range of grasp direction inputs beyond top, front, and side. Allowing the user or calling operation to specify a more detailed approach direction would make the system more adaptable and task-aware. This aligns with emerging research on task-oriented grasping, which shows that grasps designed with an understanding of the broader task lead to improved, more beneficial outcomes. However, incorporating task-oriented reasoning into a non-deep learning system presents its own difficulties as the grasp objectives must be described into clear, interpretable instructions while maintaining low latency and computational efficiency.

Finally, and perhaps most importantly, this project provides a foundation for a broader line of research aimed at bridging geometric grasping and deep learning-based manipulation. The system's ability to generate fast, low-cost initial grasp estimates makes it a strong candidate for integration into hybrid pipelines with a deep learning model. In unknown environments where learned methods struggle with unfamiliar objects, this proposed system can serve as an initial step, improving convergence, reducing failures, and enabling safer operation. Through these avenues, this work can contribute to the development of more generalisable, data-efficient, and dependable grasping systems capable of operating effectively in real-world, unstructured environments.

Chapter 6

Conclusion

This thesis set out to determine whether a superquadric-based manipulation pipeline could serve as an efficient and reliable alternative to deep learning for grasp estimation in human service robots.

The proposed three-stage framework, consisting of object decomposition into base components, adaptive superquadric fitting, and a geometric grasp selection process, demonstrated that useful 6D grasp poses can be generated with minimal computation and without prior knowledge of the grasped object. The efficiency results shown in section 4 reinforced this notion, as the system achieved the fastest runtime and the lowest memory usage among all comparison methods. Evidently, making a strong case for the approach's suitability for robots that need to operate with strict computational constraints and require real-time responsiveness.

In terms of accuracy, the first set of tests showed that the system performs well in isolated and cluttered scenes but struggles when the object is under heavy occlusion, which was expected given the system's reliance on single-view depth data. This limitation highlights an important trade-off identified throughout the project: geometric methods can offer major gains in speed and resource efficiency, but they do not yet match the accuracy of deep-learning methods. This is primarily due to a lack of contextual understanding of the target and its surroundings.

However, the second accuracy test demonstrated that the system was able to handle many of the limitations and challenges that were raised by previous superquadric-based systems, particularly when dealing with partial views, concave shapes, and irregular object geometries.

Taken together, these findings show that the proposed superquadric-based solution offers a promising approach for efficient grasp estimation, particularly in systems where a robot's resources are limited or held up by other systems. As such, there is clear potential for further development both on this project and research stemming from it. Enhancing the system with contextual information from the robot's world model and incorporating more semantic cues during segmentation can further develop the system's understanding of its environment and task. Additionally, using the pipeline as a fast initial estimate for a learning-based method could improve robustness while preserving efficiency. With continued refinement, approaches like the one presented here may contribute to grasping systems that are not only faster and more economical but also more adaptable and reliable in real-world environments.

Bibliography

[1] Superquadrics. `https://superquadrics.com/`. Accessed: 2 November 2025.

[2] Alan H Barr. Superquadrics and angle-preserving transformations. *IEEE Computer graphics and Applications*, 1(01):11–23, 1981.

[3] Laurent Chevalier. Segmentation and superquadric modeling of 3d objects. 2003.

[4] Guoguang Du, Kai Wang, Shiguo Lian, and Kaiyong Zhao. Vision-based robotic grasping from object localization, object pose estimation to grasp estimation for parallel grippers: a review. *Artificial Intelligence Review*, 54(3):1677–1734, 2021.

[5] Claire Dune, Eric Marchand, Christophe Collowet, and Christophe Leroux. Active rough shape estimation of unknown objects. In *2008 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 3622–3627. IEEE, 2008.

[6] Hao-Shu Fang, Chenxi Wang, Minghao Gou, and Cewu Lu. Graspnet-1billion: A large-scale benchmark for general object grasping. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11444–11453, 2020.

[7] Corey Goldfeder and Peter K Allen. Data-driven grasping. *Autonomous Robots*, 31(1):1–20, 2011.

[8] Grammarly Inc. Grammarly, 2025. AI assisted spelling and grammar correction tool.

[9] Stefan Hinterstoisser, Vincent Lepetit, Slobodan Ilic, Stefan Holzer, Gary Bradski, Kurt Konolige, and Nassir Navab. Model based training, detection and pose estimation of texture-less 3d objects in heavily cluttered scenes. In *Asian conference on computer vision*, pages 548–562. Springer, 2012.

[10] Tomas Hodan, Frank Michel, Eric Brachmann, Wadim Kehl, Anders GlentBuch, Dirk Kraft, Bertram Drost, Joel Vidal, Stephan Ihrke, Xenophon Zabulis, et al. Bop: Benchmark for 6d object pose estimation. In *Proceedings of the European conference on computer vision (ECCV)*, pages 19–34, 2018.

[11] Michael Krainin, Peter Henry, Xiaofeng Ren, and Dieter Fox. Manipulator and object tracking for in-hand 3d object modeling. *The International Journal of Robotics Research*, 30(11):1311–1327, 2011.

[12] Jeremy Maitin-Shepard, Marco Cusumano-Towner, Jinna Lei, and Pieter Abbeel. Cloth grasp point detection based on multiple-view geometric cues with application to robotic towel folding. In *2010 IEEE International Conference on Robotics and Automation*, pages 2308–2315. IEEE, 2010.

[13] Zoltan-Csaba Marton, Dejan Pangercic, Nico Blodow, Jonathan Kleinehellefort, and Michael Beetz. General 3d modelling of novel objects from a single view. In *2010 ieee/rsj international conference on intelligent robots and systems*, pages 3700–3705. IEEE, 2010.

[14] Rhys Newbury, Morris Gu, Lachlan Chumbley, Arsalan Mousavian, Clemens Eppner, Jürgen Leitner, Jeannette Bohg, Antonio Morales, Tamim Asfour, Danica Kragic, et al. Deep learning approaches to grasp synthesis: A review. *IEEE Transactions on Robotics*, 39(5):3994–4015, 2023.

[15] Andreas ten Pas and Robert Platt. Using geometry to detect grasps in 3d point clouds. *arXiv preprint arXiv:1501.03100*, 2015.

[16] Andreas Richtsfeld, Thomas Mörwald, Johann Prankl, Michael Zillich, and Markus Vincze. Segmentation of unknown objects in indoor environments. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 4791–4796. IEEE, 2012.

[17] Andreas Ten Pas, Marcus Gualtieri, Kate Saenko, and Robert Platt. Grasp pose detection in point clouds. *The International Journal of Robotics Research*, 36(13-14):1455–1473, 2017.

[18] Jonathan Tremblay, Thang To, Balakumar Sundaralingam, Yu Xiang, Dieter Fox, and Stan Birchfield. Deep object pose estimation for semantic robotic grasping of household objects. *arXiv preprint arXiv:1809.10790*, 2018.

[19] Xun Tu and Karthik Desingh. Superq-grasp: Superquadrics-based grasp pose estimation on larger objects for mobile-manipulation. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3361–3368. IEEE, 2025.

[20] Stephen Tyree, Jonathan Tremblay, Thang To, Jia Cheng, Terry Mosier, Jeffrey Smith, and Stan Birchfield. 6-dof pose estimation of household objects for robotic manipulation: An accessible dataset and benchmark. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 13081–13088. IEEE, 2022.

[21] Giulia Vezzani, Ugo Pattacini, Giulia Pasquale, and Lorenzo Natale. Improving superquadric modeling and grasping with prior on object shapes. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6875–6882. IEEE, 2018.

[22] Brayan S Zapata-Impata, Pablo Gil, Jorge Pomares, and Fernando Torres. Fast geometry-based computation of grasping points on three-dimensional point clouds. *International Journal of Advanced Robotic Systems*, 16(1):1729881419831846, 2019.

Appendix 1

A.1 Source Code Repository

The project source code is hosted online, along with demonstration videos and a user guide:

`https://github.com/j-kanbour/Efficient-Grasp-using-Superquadrics-for-Human-Service-Robots`

A.2 Computer Build for testing

Table A.1: Computer specifications used for all simulation and evaluation tests

Component	**Specification**
Operating System	Ubuntu 20.04
ROS Version	ROS1 Noetic
CPU	AMD Ryzen 9 5900X, 12 Core Processor
GPU	NVIDIA GeForce RTX 3060
RAM	32Gb

A.3 Augmented RANSAC algorithm

Algorithm 2 Plane guided point cloud segmentation

function DefineSegments($P, \tau_s, \theta_s, \theta_i, \tau_z$)
  $S \leftarrow \emptyset$ ▷ set of segments
  $R \leftarrow P$ ▷ remaining points
  while R has enough points **do**
    ensure normals are estimated for R
    $bestPlane \leftarrow$ none
    $bestInliers \leftarrow \emptyset$
    for a fixed number of RANSAC iterations **do**
      randomly select three points p_1, p_2, p_3 from R
      if pairwise distances between $\{p_1, p_2, p_3\} \geq \tau_s$ **and** their normals differ by at most θ_s **and** their depths differ by at most τ_z **then**
        plane $\leftarrow$ plane fitted through p_1, p_2, p_3
        inliers $\leftarrow$ all points $q \in R$ that:
        (i) lie within distance threshold of the plane
        (ii) have normals within θ_i of the plane normal
        (iii) have depth within τ_z of the seed depth
        if |inliers| > |bestInliers| **then**
          $bestPlane \leftarrow$ plane
          $bestInliers \leftarrow$ inliers
        end if
      end if
    end for
    if $bestPlane$ is none **or** |bestInliers| is below a minimum size **then**
      break
    end if
    add $bestInliers$ as a new segment to S
    remove $bestInliers$ from R
  end while
  return S
end function